

Data Visualization and Analytics in Excel

(Build dashboards that identify trends and support decisions)

Overview

This course uses an artifact sequence approach in which participants build a management dashboard from operational data across five topic blocks. The dataset is imported and named before the session begins. Participants then extend the Table with calculated columns, build a multi-period summary for trend comparison, construct a Pivot Table dashboard with charts and slicers, design it for decision support with KPI indicators and conditional formatting, and finalize it as a protected, distribution-ready report.

Objectives

Upon completion of the course, participants will be able to build a dynamic, decision-ready Excel dashboard that summarizes operational data, identifies trends, and flags performance gaps for management reporting.

Specifically, participants will be able to:

- organize and extend an imported Excel Table with calculated columns and named ranges
- aggregate and compare data across categories and periods using SUMIFS and COUNTIFS
- build a Pivot Table dashboard with slicers, data validation, and a trend line chart
- design the dashboard for clarity with KPI indicators and conditional formatting
- validate and protect the completed dashboard before distribution

Who Should Participate

- Rank-and-File to Supervisory staff who prepare or present operational reports.
- Employees who need to turn raw data into visual summaries for management.
- Staff who currently report using static tables and want to build interactive dashboards.

Key Topics

- I. Structuring Data for Analysis
 - a. Extending the Table with calculated columns
 - b. Using structured Table references in formulas
 - c. Applying named ranges for reusable inputs
 - d. Organizing worksheets for a navigable workbook
- II. Aggregating and Comparing Data
 - a. Using SUMIFS and COUNTIFS to summarize data
 - b. Building a multi-period summary for trends
 - c. Using IF and IFS to classify by threshold
 - d. Using IFERROR to handle missing or error values
- III. Building the Dashboard
 - a. Creating a Pivot Table via Insert → PivotTable
 - b. Defining a named range for data validation
 - c. Adding Slicers to filter by category or period
 - d. Inserting a line chart to show trends over time
- IV. Designing for Clarity and Decision Support
 - a. Displaying KPI values with formula-driven cells
 - b. Applying conditional formatting for KPI alerts
 - c. Arranging charts, slicers, and KPIs on a sheet
 - d. Applying print settings for report distribution
- V. Validating and Finalizing the Dashboard
 - a. Testing the dashboard by refreshing the query
 - b. Using Evaluate Formula to check formula logic
 - c. Resolving #N/A, #REF!, #DIV/0!, #VALUE! errors
 - d. Protecting the sheet via Review → Protect Sheet